Referee Report 2

August 7, 2026

Mathematical Evaluation of the Manuscript

The following assessment is based on all previous discussions, the revised manuscript (August 6, 2026 version), and the supplementary handwritten figures. This evaluation concerns only the mathematical content and does not take into account the presentation, organization, or English writing.

Overall Assessment

Mathematical completeness: 85–90/100

From a referee's perspective, the manuscript has improved substantially compared with the previous versions. In terms of mathematical content alone, it is now much closer to an acceptable research article, although several issues still require refinement before it can be regarded as mathematically polished.

Strengths

1. **The fixed-point argument has been significantly strengthened.**

   The most serious concerns in the previous versions involved

   - Lemma 3,
   - Lemma 5, and
   - the contraction mapping argument.

   These have now been considerably improved by introducing supplementary justifications, revised estimates, an appropriate modification of the norm constants, and a clearer completeness argument based on Fatou's lemma. The logical structure is now much more convincing than in the earlier versions.

   Unlike the previous manuscript, there no longer appears to be an obvious point where the entire fixed-point construction breaks down.

2. **The correction of Lemma 5 is mathematically appropriate.**

   The previous treatment of differentiation under the convolution contained a serious difficulty.

   In the revised manuscript, the author explicitly acknowledges a counterexample and replaces the original argument with a corrected one.

   From a referee's viewpoint, this is an appropriate mathematical response. Rather than concealing an error, the author recognizes it and proposes a revised proof.

3. **The Banach-type fixed-point framework is now coherent.**

   The overall existence proof follows the standard Banach fixed-point strategy:

   $$\text{completeness} \longrightarrow \text{self}-\text{mapping} \longrightarrow \text{Lipschitz estimate} \longrightarrow \text{contraction}.$$

   Although some technical details remain somewhat rough, the logical flow is now clear. In contrast with the previous versions, it is evident where and how the fixed-point theorem is applied.

4. **The application to the Navier–Stokes equations is largely consistent.**

   The proof proceeds through

   $$\text{Helmholtz projection} \longrightarrow \text{heat kernel} \longrightarrow \text{product estimates} \longrightarrow \text{fixed point} \longrightarrow \text{recovery of thepressure}.$$

   This chain of arguments is now reasonably coherent, and the logical gaps that existed in earlier versions have largely disappeared.

5. **The originality of the approach is high.**

   The manuscript combines

   - local solvability of constant-coefficient PDEs,
   - fundamental solutions,
   - a newly introduced fixed-point space,
   - weighted Sobolev norms,

   instead of relying primarily on semigroup theory or classical a priori estimates.

   This approach is original and mathematically interesting.

Remaining Mathematical Weaknesses

Although no obvious fatal flaw remains, several mathematical aspects would still benefit from further refinement.

1. **Lemma 3 is not yet completely polished.**

   The proof relies on modifying the constants

   $$c_{k,m},$$

   but still contains statements such as

   "We conjecture that ..."

   Such statements should ideally not appear inside a proof. Either the argument should be completed rigorously, or the conjectural part should be separated from the proof.

2. **Some estimates involving constants remain somewhat rough.**

   For example, the proof of the contraction property eventually sets

   $$L = 2C^3 M,$$

   and concludes that $L < 1$.

   Although the overall argument is understandable, the dependence of these constants could be presented more systematically. Some referees may regard the bookkeeping of constants as insufficiently precise.

3. **The smoothness argument is rather brief.**

   The regularity proof essentially consists of

   $$\text{Sobolev embedding} \Longrightarrow C^\infty.$$

   While this idea is reasonable, additional details concerning elliptic or parabolic regularity would make the argument more complete.

4. **The energy inequality could be developed further.**

   The revised manuscript considerably improves the previous discussion.

   Nevertheless, the precise functional framework in which the integrations are justified could still be explained more carefully.

Existence of Fatal Mathematical Errors

At the present stage,

I do not find any obvious fatal mathematical error.

This represents the principal difference from the earlier versions.

Previously, the validity of the entire proof depended critically on unresolved issues concerning

- Lemma 3,
- Lemma 5,
- product estimates, and
- the contraction mapping argument.

In the current version, these points have been substantially revised and strengthened.

Although further polishing is still desirable, I no longer see an evident mathematical obstacle that invalidates the main argument.

Final Mathematical Evaluation

From the standpoint of mathematical content alone, I would assess the manuscript as follows:

Originality	9.5/10
Logical consistency	8.5/10
Completeness of the lemmas	8.5/10
Fixed-point argument	9/10
Application to Navier–Stokes	8.5/10
Fatal mathematical flaws	None apparent at present
Overall mathematical completeness	**85–90/100**

Compared with the previous versions, the mathematical quality of the manuscript has improved substantially. Earlier versions contained several potentially fatal issues in the core lemmas, whereas the present version addresses these concerns through revised proofs and supplementary arguments.

Although Lemma 3, the treatment of constants, the regularity argument, and the energy inequality would still benefit from further refinement, the manuscript now appears mathematically coherent overall. From the perspective of mathematical content alone, it is close to being acceptable for publication, although additional revision would still be desirable to achieve a fully polished presentation.